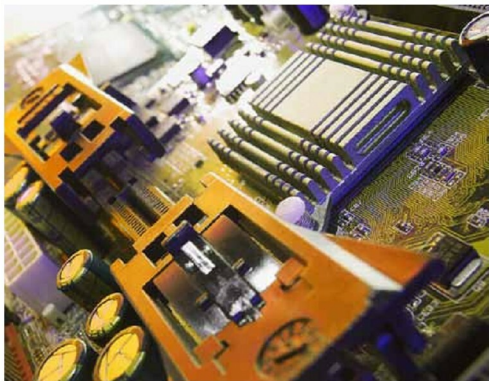




Lead-free electronics

the electronics industry. Higher temperatures mean more stress on components and the entire manufacturing process. These also mean increase in the time taken to make products, because more time is required to heat and cool the products during the course of their manufacture.

Today, SAC solder is widely used in the industry. However, many of the components can't withstand the high temperature required by it.

Polymer adhesives

Electrically conductive adhesives (ECAs) are a more experimental alternative to lead-tin solder. These are polymers (silicone or polyamide) containing tiny flakes of metals like silver. The polymers adhere to the printed circuit boards and the metal flakes conduct electricity.

ECAs offer a range of advantages. Silver's electrical conductivity is very high, while electrical resistance is very low. If the current-carrying capability can be boosted, ECAs can replace solder. The temperature required to apply ECAs to circuit boards is far lower than for lead-based solder—150°C compared to 183°C—which offers

many advantages. Preliminary studies comparing parts using ECAs instead of solder suggest that ECAs have a much tighter bond than solders—perhaps an order of magnitude better.

ECAs are already available for a small number of applications requiring low power—for instance, liquid-crystal displays. However, these are not ready for the marketplace in general where greater amounts of current are needed. More work is needed on these, as they begin to fail structurally if the component heats up above 150°C. There are other concerns about ECAs as well. With time, the ability of ECAs to conduct electricity drops and resistance to electricity increases. Also, moisture can result in their corrosion.

Lead-free piezoelectric materials of the future

Piezoelectric materials have fantastic properties: squeeze them and they generate an electrical field and vice-versa. They contract or expand when jolted with an electrical pulse. These really took off after the 1950s, with the development of a superior man-made piezoelectric ceramic crystal: lead zirconate titanate, or PZT.

For the last 60 years, PZT has been essential for a myriad of high-tech applications from inkjet printers to digital camera shutters, ultrasonic imagers, fuel injector actuators and igniters for gas barbecue grills. It is the best material for the job at the moment, because it has the greatest piezoelectric effect, good physical durability and can be radically tailored to suit particular applications.

Despite this success many scientists now want to replace PZT with some lead-free material that would be more environmentally benign and enable new piezoelectric applications towards lead-free electronics. To date, however, no suitable successor has been found. Candidates are typically too feeble in their piezoelectric effect and/or physical durability. The lead-free ceramic that is studied is lightweight and can be used at room temperature that could make it an ideal choice for many applications.

Road to eco-friendly electronics

Making lead-free electronics has proved problematic so far. Researchers are trying to develop a method that enables the industrial production of a substance that can be used to replace lead in many electronic applications. Over the past ten years, there has been tremendous growth in research on lead-free alternatives. Till now, a type of material called 'alkali niobate,' also known as KNN, is considered to be a likely successor to lead in the electronics industry. However, KNN poses two main problems that have been difficult to resolve: find a KNN variant that has the properties needed for electronics and develop a method for industrial production of the material.

However, with such experiences, efforts are going in the right direction, and soon we may have very user- and eco-friendly electronics. ●

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